

## QT 405: Introduction to Quantum Computation

**Course Code:** QT 405

**Course Name:** Introduction to Quantum Computation

**L-T-P-C:** 3-0-0-3

**Prerequisites:** Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model

**Intended for:** UG/PG/PhD

**Mutual Exclusion:** CS 520, PH 579

**Distribution:** Elective PG/Elective UG

**Semester:** Odd/Even

**Approval:** 47th Senate

### Preamble

Students of this course learn (i) To review the basic postulates of quantum mechanics, (ii) The theoretical basics of qubits and their physical realisations, (iii) To work with density operators and time evolution for mixed states, (iv) The basic ideas of quantum gates, (v) The working of important quantum algorithms, (vi) The basics of quantum error correction

### Course Content and syllabus

- **Qubits versus classical bits**
  - Spin-half systems and photon polarizations
  - Trapped atoms and ions
  - Artificial atoms using circuits
  - Semiconducting quantum dots
  - Single and Two qubit gates – Solovay - Kitaev Theorem
- **Quantum correlations**
  - Entanglement and Bell's theorems
- **Review of Turing machines and classical computational complexity**
  - Time and space complexity (P, NP, PSPACE)
- **Reversible computation**
- **Universal quantum logic gates and circuits**
- **Quantum algorithms**
  - Deutsch algorithm
  - Deutsch Josza algorithm
  - Bernstein - Vazirani algorithm
  - Simon's algorithm

- **Database search**
  - Grover's algorithm
- **Quantum Fourier Transform and prime factorization**
  - Shor's Algorithm.
- **Quantum complexity classes – Q, EQP, BQP, BPP, QMA**
- **Additional Topics in Quantum Algorithms**
  - Variational Quantum Eigensolver (VQE)
  - HHL
  - QAOA
- **Introduction to Error correction**
  - Fault-tolerance
  - Simple error correcting codes
- **Survey of current status**
  - NISQ era processors
  - Quantum advantage claims
  - Roadmap for future

## Course References

1. *Quantum Information Science* – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)
2. *Quantum computation and quantum information* – Nielsen M. A., and Chuang I. L., 10th Anniversary edition, Cambridge University Press (2010)
3. A Pathak, *Elements of Quantum Computation and Quantum Communication*, Boca Raton, CRC Press (2015)
4. *Quantum error correction and Fault tolerant computing*, Frank Gaitan, 1st edition, CRC Press (2008)
5. *Quantum computing explained*, David McMahon, Wiley (2008)
6. *Introduction to Quantum Computing: From a lay person to a programmer in 30 steps*, Hui Yung Wong, 1st edition, Springer-Nature Switzerland AG (2022)